

Motor Datasheet: tm SAR
20260127 165414 v2b 012Nm
okLoss v2 20C 2026-07-20
13-42-54

Operating Point: Continuous | Temperature: 20 °C

Geometry (active elements only)

Parameter	Value	Unit
Machine Outer Diameter	55.0	mm
Machine Inner Diameter	20.0	mm
Machine Thickness / Width	23.7	mm
Stator Height	23.7	mm
Rotor Height	23.7	mm
Magnet Height	3.7	mm
Tooth Height	13.9	mm
Air Gap	0.50	mm
Fill Factor (input)	0.96	-
Slot Fill Factor (calculated)	0.78	-
Slot Width (tooth to tooth)	4.292	mm
Available Coil Space	2.146	mm
Assumed Coil Width (simulation)	1.946	mm
Assumed Coil Height (simulation)	12.356	mm

Mass Properties (active elements only)

Parameter	Value	Unit
Machine Mass	299.3	g
Rotor Mass	60.3	g

Stator Mass	239.0	g
Total Copper Mass (pure copper)	110.2	g
Total Stator Steel Mass	128.7	g
Total Magnet Mass	26.2	g
Rotor Structure Included	No	-
Rotor Moment of Inertia (active elements only)	245.8	g·cm ²

Electrical Properties

Parameter	Value	Unit
Supply Current (RMS)	5.213	A
Supply Current (Peak)	7.373	A
Id (Peak)	-0.000	A
Iq (Peak)	7.373	A
Phase Advance Angle	0.0	°elec
Phase Resistance	47.098	mOhm
D-Axis Inductance (Ld)	197.071	μH
Q-Axis Inductance (Lq)	196.779	μH
Terminal Inductance (Ltt = Ld + Lq)	393.850	μH
PM Flux Linkage	2.771	mWb
Frequency at operating point (electrical)	237.500	Hz
Winding Configuration	wye (star)	-
DC Supply Voltage	9.000	V
Power Factor	0.901	-

Performance

Parameter	Value	Unit
Torque (Continuous)	152.5	mNm
Speed	2850.0	rpm
Dimensionless motor constant	4.8	-
BackEMF Constant (Ke, phase peak)	1.5	mV/rpm
Motor Constant (Km, only resistive losses)	77.8	mNm/√W
Specific Motor Constant (Km / mass)	260.1	mNm/√W/kg
Torque Constant (Kt, peak, no-load)	20.8	mNm/A

Torque Ripple	1.6	%
Saturation	0.5	%

Losses

Parameter	Value	Unit
Total Losses	4.5	W
Resistive Losses	3.8	W
Iron Core Losses	0.6	W
Magnet Eddy Current Losses	0.0	W
Rotor Structure Eddy Current Losses	0.0	W
Winding Eddy Current Losses	0.0	W

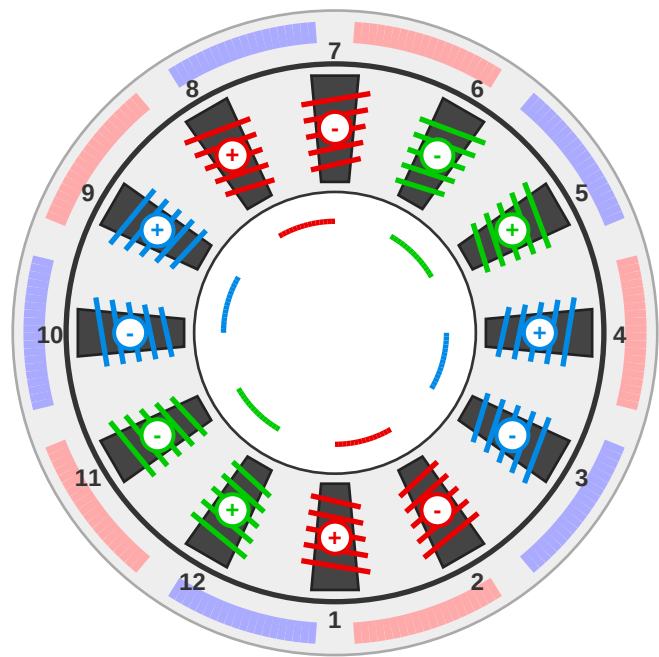
Configuration

Parameter	Value	Unit
Number of Stators	1	-
Number of Rotors	1	-
Number of Slots	12	-
Pole Pairs	5	-
Number of Parallel Paths	1	-
Number of Turns per Tooth (designed incl. parallel paths)	19.0	-
Number of Phase Turns (all phases in series)	76.0	-

Winding Configuration

Schema: **AabBCcaABbcC**

Concentrated Winding | Balanced | Cogging: 60/rev | Ripple: 60/rev



Materials

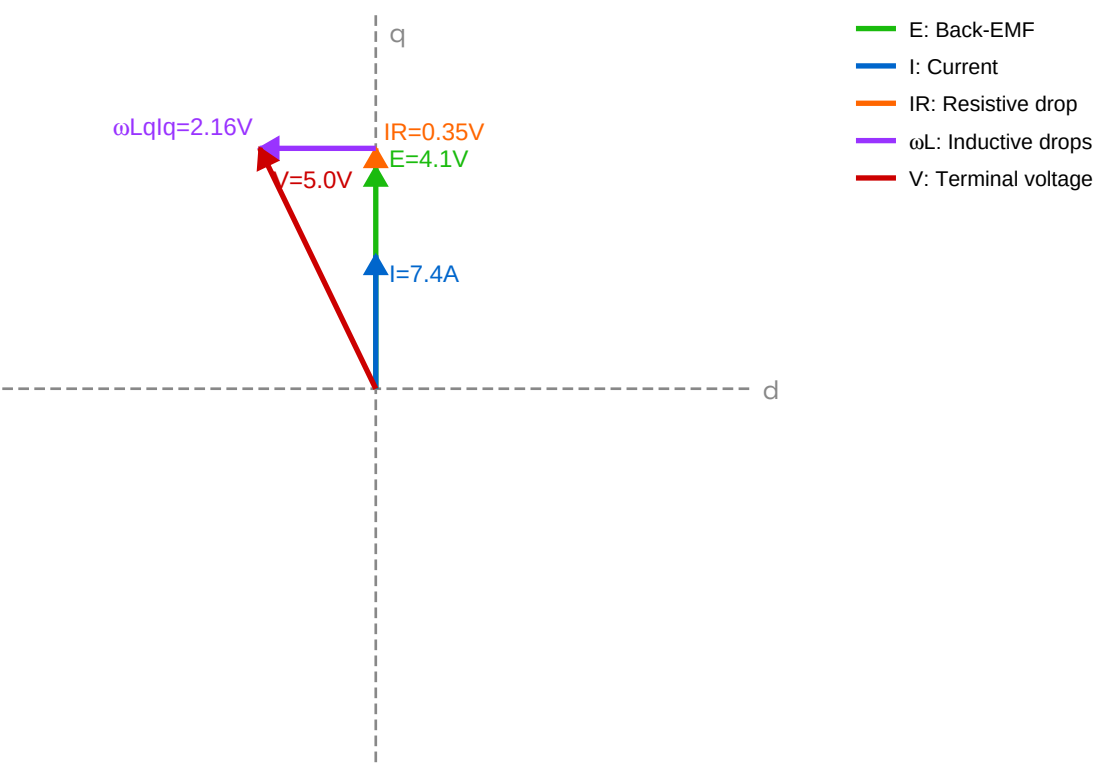
Component	Material	
Stator Core	700HR 5P 800MPa-Hoganas	-
Magnets	BMHFa-32/27	-
Rotor Backiron	430-FR 15mm-Carpenter	-
Coil	Copper	-

Voltage Breakdown at Operating Point (with integer turns)

Parameter	Value	Unit
Back-EMF Voltage (Phase, Peak)	4.136	V
Resistive Drop (Phase, Peak)	0.347	V
Inductive Drop Ld (Peak)	0.000	V
Inductive Drop Lq (Peak)	2.165	V

Inductive Drop Total (Peak)	2.165	V
Terminal Voltage (Phase, Peak)	4.976	V
Back-EMF Voltage (Line-to-Line, Peak)	7.164	V
Resistive Drop (Line-to-Line, Peak)	0.601	V
Inductive Drop (Line-to-Line, Peak)	3.750	V
Terminal Voltage (Line-to-Line, Peak)	8.619	V

Phasor Diagram Phase Values NOT Line-to-Line (d-q reference frame)



Mechanical Properties

Parameter	Value	Unit
Airgap Axial Force (Average)	39.34	N
Magnet Axial Force Positive Maximum	0.00	N
Magnet Axial Force Positive Minimum	0.00	N
Magnet Axial Force Negative Maximum	-15.37	N
Magnet Axial Force Negative Minimum	-10.93	N

Glossary

- Peak — Refers to 0-to-peak amplitude (NOT peak-to-peak). For sinusoidal waveforms: $\text{Peak} = \text{RMS} \times \sqrt{2}$.
- RMS — Root Mean Square, the equivalent DC value for power calculations.
- Line-to-Line (L-L) — Voltage measured between two phases. For wye connection: $V_{LL} = V_{\text{phase}} \times \sqrt{3}$.